

Project Schedule and Resource Planning

"The following schedule outlines the planned development activities, estimated duration, project milestones, resource allocation, and potential project risks associated with the Machine Learning—Based Credit Card Fraud Prediction System. The plan ensures systematic execution, efficient resource utilization, timely completion, and successful deployment of the proposed solution.

1. Project Timeline

		Estinuted Duration	Deliverable
1	Requilenent Amulysis	i,Væk	Softwme Requitenrnt Specification (SRS)
2	lilta•æt Collætion and Analysis	Week	Prepared and Verified lilta<
3	[ilta He-processing and Feature Fnginæring	1 Week	Cleaned and Processed Dataset
4	Machine IÄning Mo&l Development	2 Weeks	"Jiained and Evaluated Pædiction Mo&ls
5	Flask Web AA)11lcation Development	2 Weeks	Functional YVeb Application
6	System Testing and Validatiq:mi		Testing and Validation Report
7	L'kployment and Lix:umentation	1 Week	I&Dloyed Application and final PIOject Documentation

2. Major Milestones

Milestone	Expected Outcome
Requirement Analysis Completed	Approved Project Requirements
Dataset Prepared	Clean Dataset Ready for Training
Machine Learning Completed	Best-performing Model Selected
Web Application Developed	Theoretical System Successfully Integrated
System Testing Completed	Verified Functional System
Deployment Completed	Application Successfully Hosted and Ready for Demonstration

3. Risk Assessment

	Mitigation
Poor Data Quality	thorough preprocessing, validation, and data cleaning.
Low Prediction Accuracy	Train multiple Machine Learning algorithms and select the best-performing one.
Deployment Failure	Test deployment thoroughly using Docker containers before production deployment on Kubernetes.
Security Vulnerabilities	Implement secure authentication, encrypted communication, and protected database access.
Project Schedule Delays	Monitor milestones regularly and distribute development tasks efficiently among team members.

4. ReM»urve Allocation

Resource	Technology	Purpose
	Python 3.x	Machine Learning development and backend programming.
Backend Framework		REST endpoints, business logic, authentication, and prediction
Frontend	HTML5, CSS3, JavaScript, Bootstrap 5	Builds a responsive and user-friendly web interface,
	MySQL	Stores applicant information, user accounts, application and prediction history securely.
Machine Learning Libraries	Scikit-learn, Pandas, NumPy	Supports data preprocessing, feature engineering, training, evaluation, and prediction.
Development Tools	Visual Studio Code, Jupyter Notebook; Postman	Provides application development, machine learning experimentation, REST API serialization, and API testing-
Control	Git and GitHub	Manages source code, project collaboration, version tracking, and repository maintenance.

Cloud		Hosts the Flask web application with IVIahle inti-astructuæ and
Containerization		Packages the application into portable containers for consistent developrrw:nt, testing, and deployment across environrnents,,
Conclusion project planning strategy provides a structured Irmwlmap for successfully innplen•ænting the rvlachine l.carning—Based Cxedit Card Appaoval Prediction System. By &fining project phases, milestones, allocation, deployment strategy, and risk timelines, 1 management practices, the project efficient ct ensures &ælopnænt, ævurate pnxiiiction Irrfcmmnance, data handling, and reliable cloud CEploynænt.		